

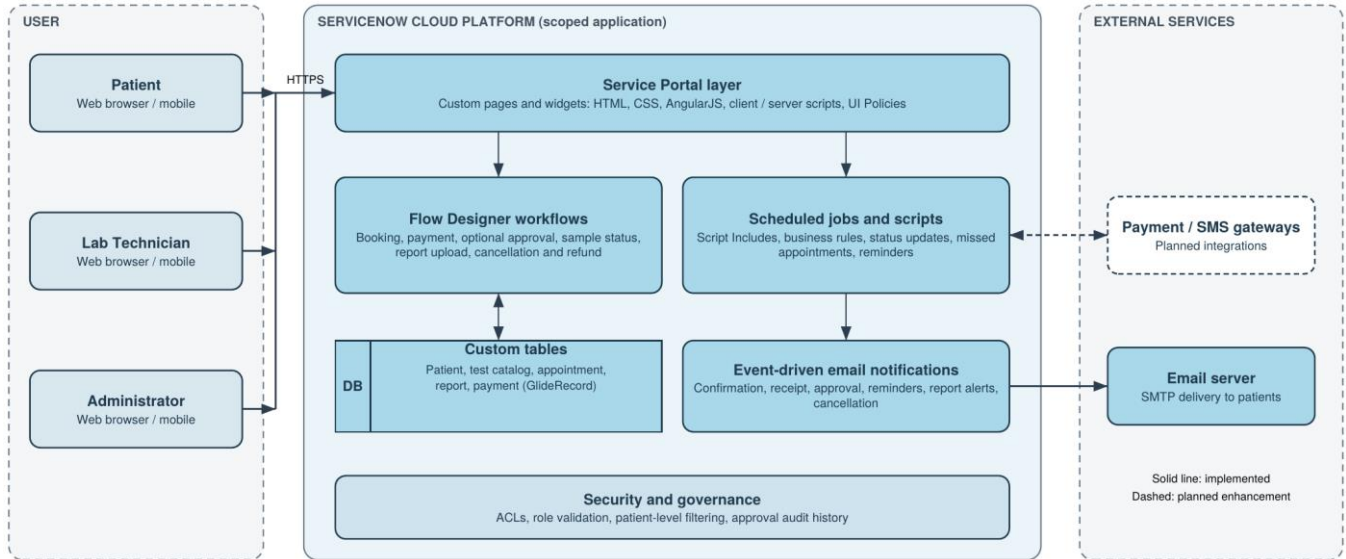
Project Design Phase-II Technology Stack (Architecture & Stack)

Date	07 October 2026
Team ID	
Project Name	Diagnostic Test Center Portal
Maximum Marks	4 Marks

Technical Architecture:

The deliverable includes the architectural diagram below and the information as per Table-1 and Table-2.

Example: Online diagnostic test booking and report management on ServiceNow



Architecture notes:

- Processes: portal booking, slot validation and pricing, Flow Designer workflow with approvals, scheduled jobs and notifications are shown as application logic blocks.
- Infrastructure: the whole solution runs in the ServiceNow cloud; users access it through a browser (no local server).
- External interfaces: email server today; payment and SMS gateways planned (dashed).
- Data storage: custom ServiceNow tables with attachments for reports. Machine learning model: not applicable.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Service Portal pages and custom widgets for test selection, slot availability, booking summary and cost, appointment tracking and report download; UI Policies and client scripts control dynamic form behavior.	HTML, CSS, AngularJS, Client / Server Scripts, UI Policies
2	Application Logic-1	Booking and slot logic: validates slot availability, prevents overlapping appointments and auto-calculates test charges.	Flow Designer, Script Includes, GlideRecord (server scripts)
3	Application Logic-2	Payment, approval and cancellation logic: payment confirmation, role-based approvals (Admin / Lab Manager), conditional approval for high-value tests, cancellation penalties and refund handling.	Flow Designer, scripted actions, Approvals
4	Application Logic-3	Automation and communication: scheduled jobs update status after report upload, mark missed appointments and send reminders; event-driven emails for confirmations, receipts and report alerts.	Scheduled Jobs, Events, Notifications, dynamic email scripts
5	Database	Custom tables for patient details, diagnostic test catalog, appointments, reports and payment transactions. Relationships: Patient to Appointment (one-to-many), Appointment to Report (one-to-one), Appointment to Diagnostic Test (many-to-one).	ServiceNow custom tables, reference fields, GlideRecord
6	Cloud Database	Tables are hosted in the database of the ServiceNow cloud instance; no separate database server is managed.	ServiceNow Cloud instance database
7	File Storage	Diagnostic reports uploaded by lab technicians are attached to report records and downloaded by patients through the portal.	ServiceNow attachments (platform storage)
8	External API-1	Email delivery of appointment confirmation, payment receipt, approval updates, reminders, report alerts and cancellation notices.	Email notifications (SMTP email service)
9	External API-2	Planned integration with external payment gateways and SMS notifications (future enhancement).	Payment gateway API, SMS gateway (planned)
10	Machine Learning Model	Not used in the current version; analytics expansion is planned for the future.	Not applicable
11	Infrastructure (Server / Cloud)	Scoped application deployed on the ServiceNow cloud platform. Local server configuration: not required, access is through a web browser. Cloud server configuration: ServiceNow instance hosting portal, workflows, tables and jobs.	ServiceNow Cloud, scoped application

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Front-end widgets are built on frameworks bundled with Service Portal.	AngularJS, Bootstrap
2	Security Implementations	ACL rules restrict read, write, create and delete access, report upload and payment modification; role validation in widget server scripts; patient-level filtering so users see only their own appointments and reports; sensitive patient and payment data protected in line with healthcare privacy standards; approval audit history.	ServiceNow ACLs, roles, Debug Security Rules
3	Scalable Architecture	Three-tier design (Service Portal, logic layer, data layer) built as a scoped application with modular widgets, reusable Script Includes, configurable Flow Designer actions and structured table relationships; ready for payment gateway, SMS, doctor portal and analytics extensions.	Service Portal, Flow Designer, Script Includes, scoped app
4	Availability	Runs on the managed ServiceNow cloud platform; background jobs run on a schedule; Flow Execution Details and system logs help detect and fix failed actions quickly.	ServiceNow Cloud, Flow Execution Details, System Logs
5	Performance	Server-side GlideRecord queries, real-time slot validation, reusable components and minimal hardcoding; reminders and status updates run as background jobs so users are not delayed.	GlideRecord, Scheduled Jobs, Service Portal preview tools

Monitoring and Maintainability:

- Monitoring: Flow Execution Details, System Logs, gs.info() debugging, GlideRecord trace analysis, Service Portal preview and Debug Security Rules validate data flow, automation triggers and role-based access.
- Best practices: scoped application development, proper naming conventions, role-based security design, minimal hardcoding and structured table relationships.

References:

<https://developer.servicenow.com/>

<https://c4model.com/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>